

Act II: Casting the Buyers

Motivations and Choices of the Citizens

Module IV: The Shopping List

Demand Curve, Market Demand, Demand Shifters



Previously in Marginal Falls

- ◆ **Scarcity** forces choices.
- ◆ **Opportunity cost** tells us what must be given up.
- ◆ **Marginal utility** tells us what one more unit adds.
- ◆ **Budget constraints** tell us what is affordable.
- ◆ A **price change** changes the best bundle.

Next: From Individual Choice to Market Demand

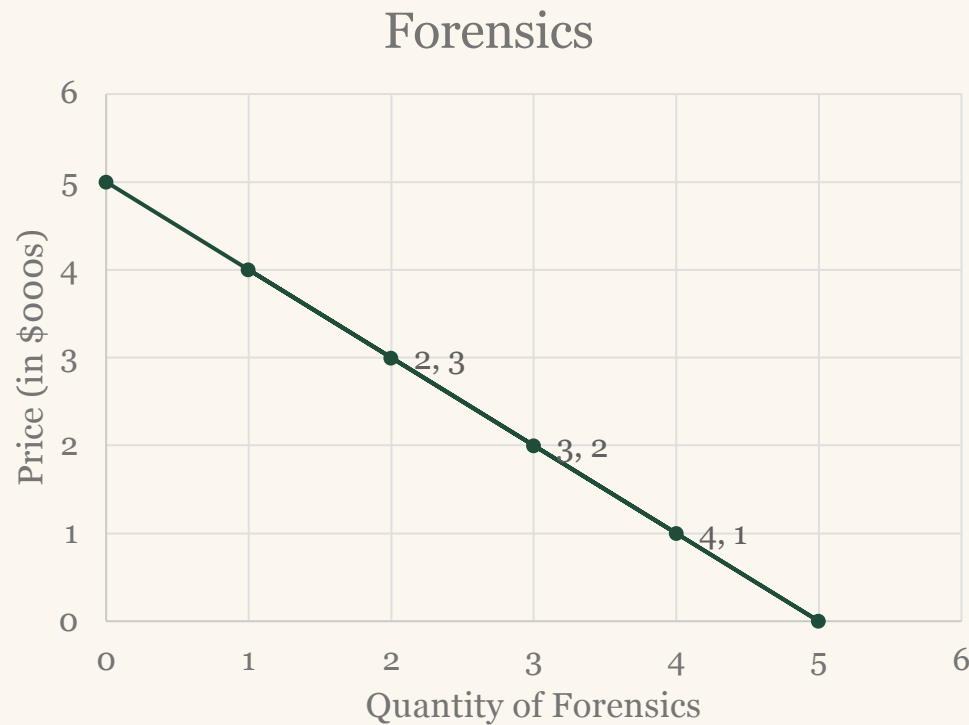
Back to the Detective Department

- Last class: The Chief had to choose between Detectives and Forensics.
- When the price of forensics changed, the best bundle changed.
- That gave us the Chief's demand for forensics.

Price of Forensics	Forensics	Detectives	Conviction Rate
\$1000	4	6	100%
\$2000	3	4	87%
\$3000	2	4	75%

The Chief's Demand for Forensics

At each price, you gave the Chief a contingency plan with the quantities.

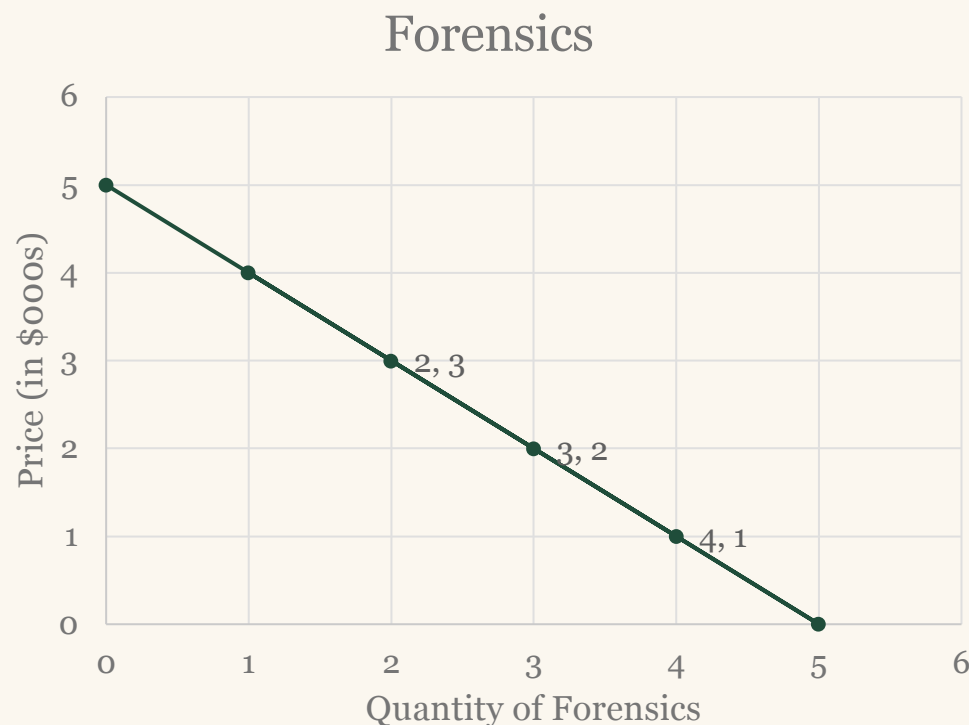


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The Chief's Demand for Forensics

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Those quantities are the **quantity demanded**.



Price of Forensics	Forensics	Detectives	Conviction Rate
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Quantity Demanded

Quantity demanded is the amount of a good or service that a buyer is willing and able to purchase at a given price.

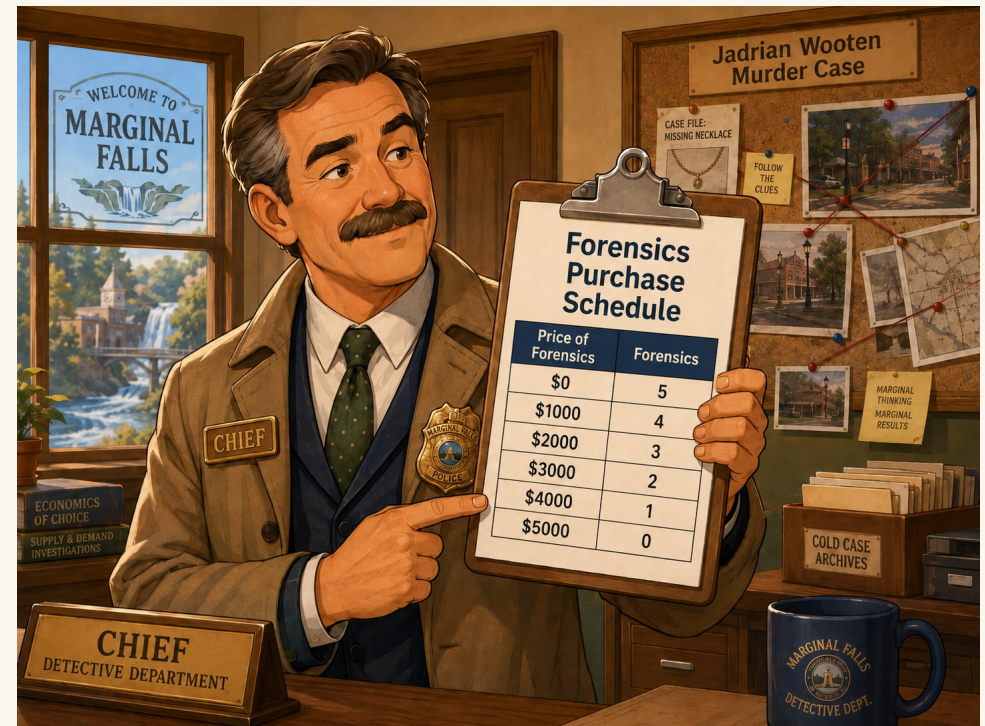
Price of Forensics	Forensics	Detectives	Conviction Rate
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At \$2,000 per unit, the Chief demands 3 units of forensics.

Demand Schedule

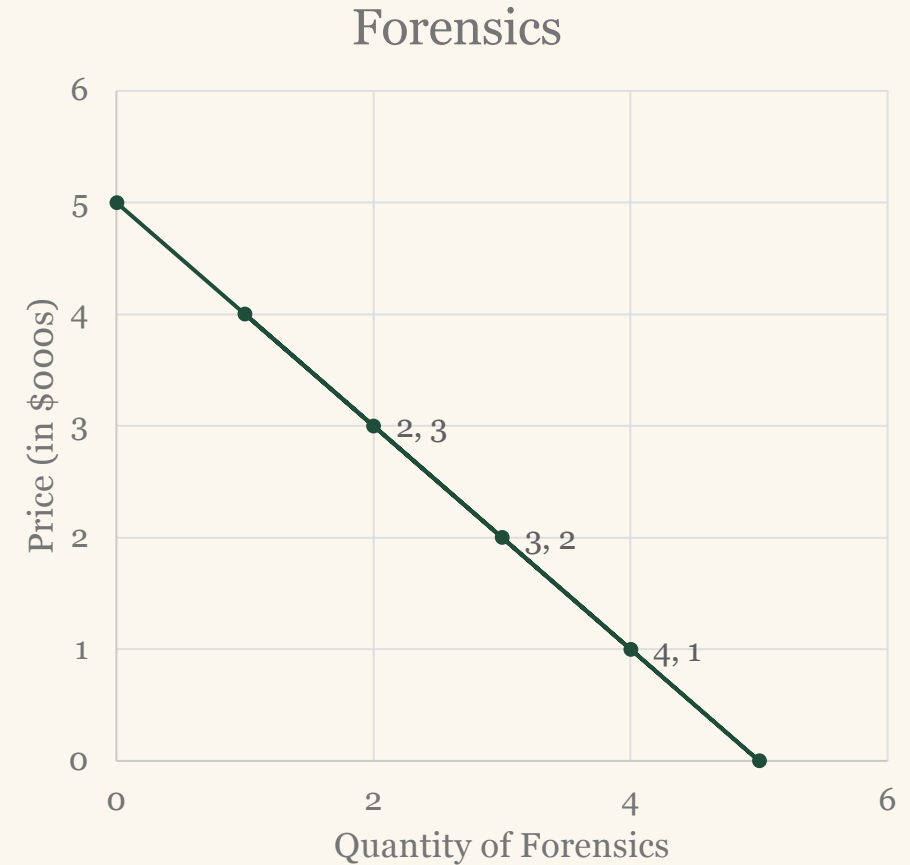
A demand schedule is a table showing the relationship between price and quantity demanded.

Demand Schedule	
Price of Forensics	Forensics
\$0	5
\$1000	4
\$2000	3
\$3000	2
\$4000	1
\$5000	0



Demand Curve

A demand curve shows the relationship between the price of a good and the quantity demanded, *holding everything else constant*.



Ceteris Paribus

- Ceteris paribus = “holding everything else constant.”
- When drawing demand, we change only the price of the good.
- We hold fixed: income, tastes, related prices, expectations, number of buyers.

In the contingency plan, the price of forensics changes from \$1,000 → \$2,000 → \$3,000. The budget, detective price, the murder case, and conviction rate calculation stay the same.

Law of Demand

Holding everything else constant:

- When price rises, quantity demanded falls.
- When price falls, quantity demanded rises.



Forensics price rises from \$1,000 to \$3,000.
Quantity demanded falls from 4 to 2.

Why Does Demand Slope Downward?

- Last class, we saw what happens when forensics becomes more expensive.
- Two things happen at once:
 - Forensics becomes relatively expensive. **(Substitution Effect)**
 - The same budget buys less evidence. **(Income Effect)**

This is why a higher price leads to a lower quantity demanded.

The Chief's Call

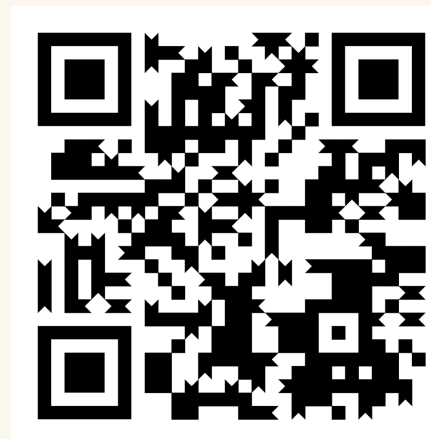
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Part I:

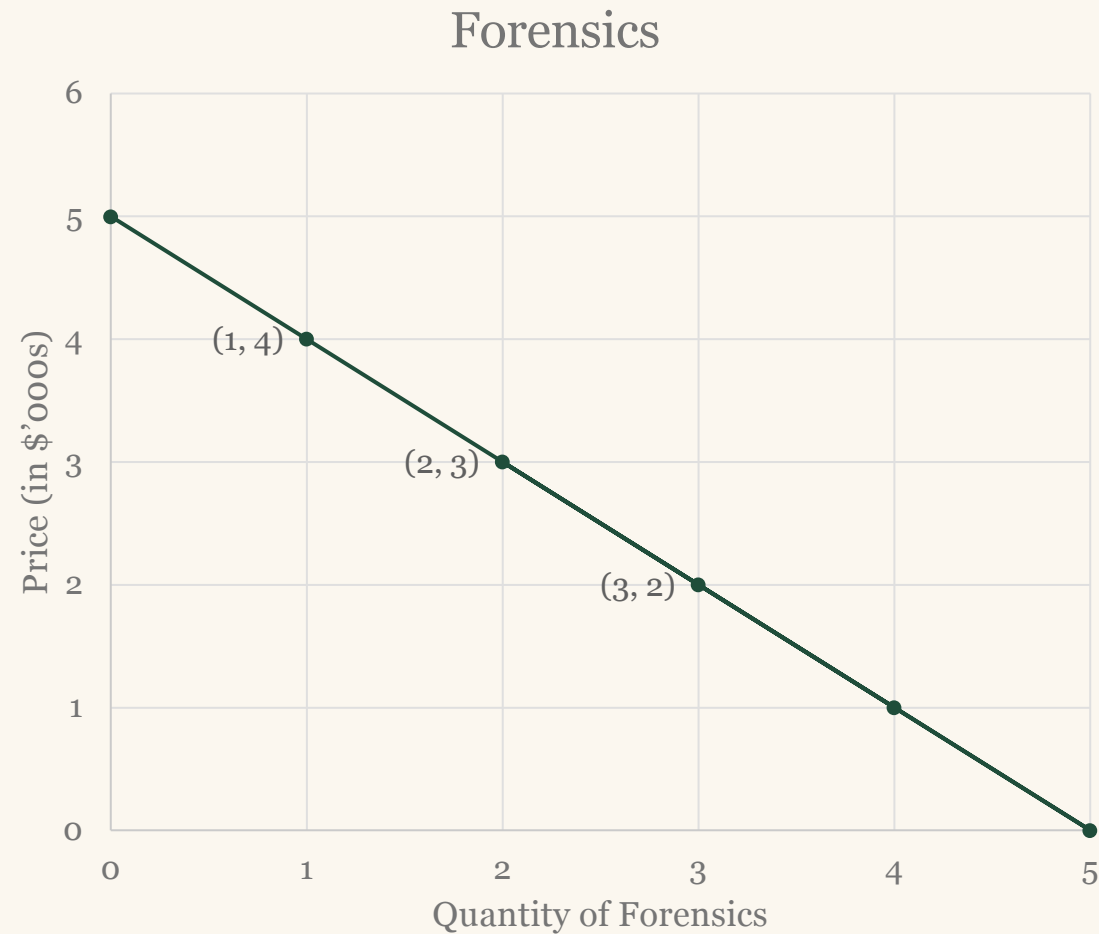
- The department now has a larger investigation budget.
- Forensics prices still vary: \$1,000, \$2,000, \$3,000.
- Detectives still cost \$1,000 each.

Part II:

- Now detectives become more expensive.
- Forensics prices still vary.

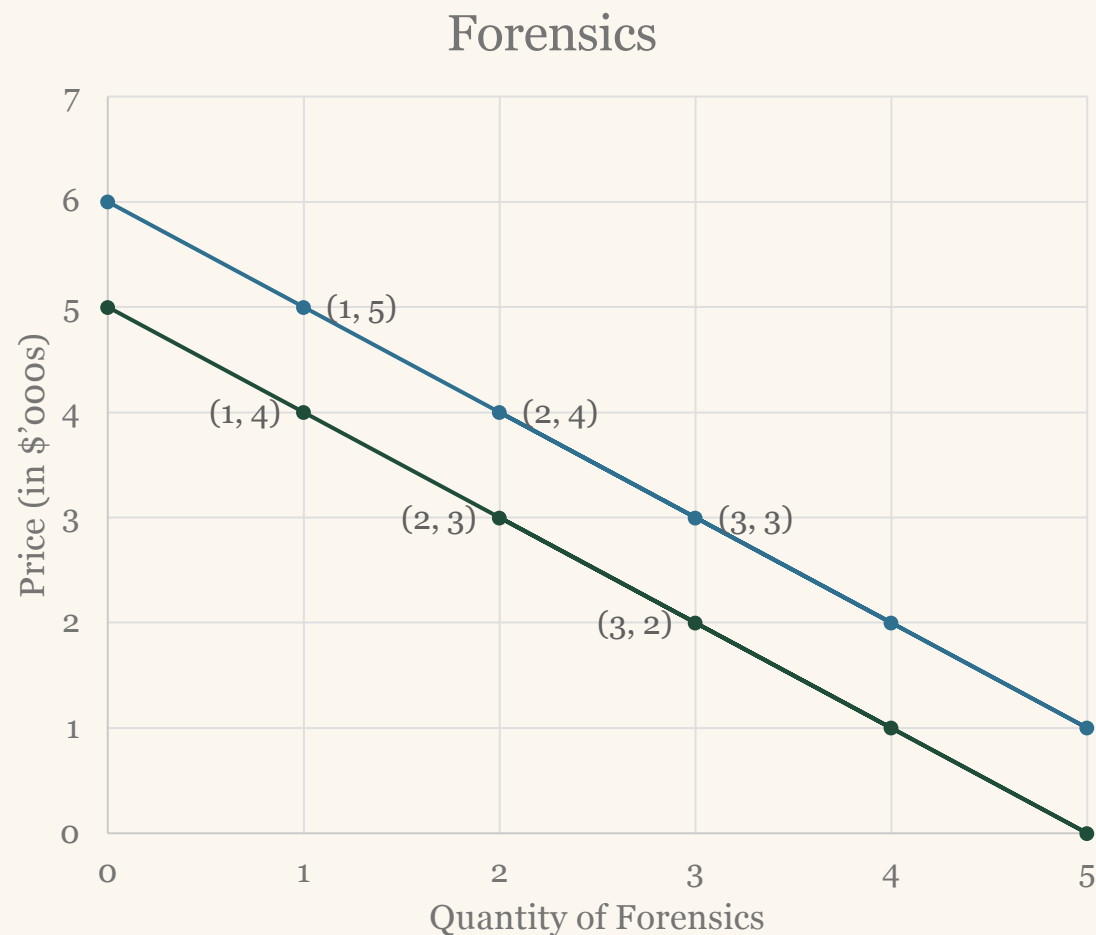


What Happened When Budget Increased?



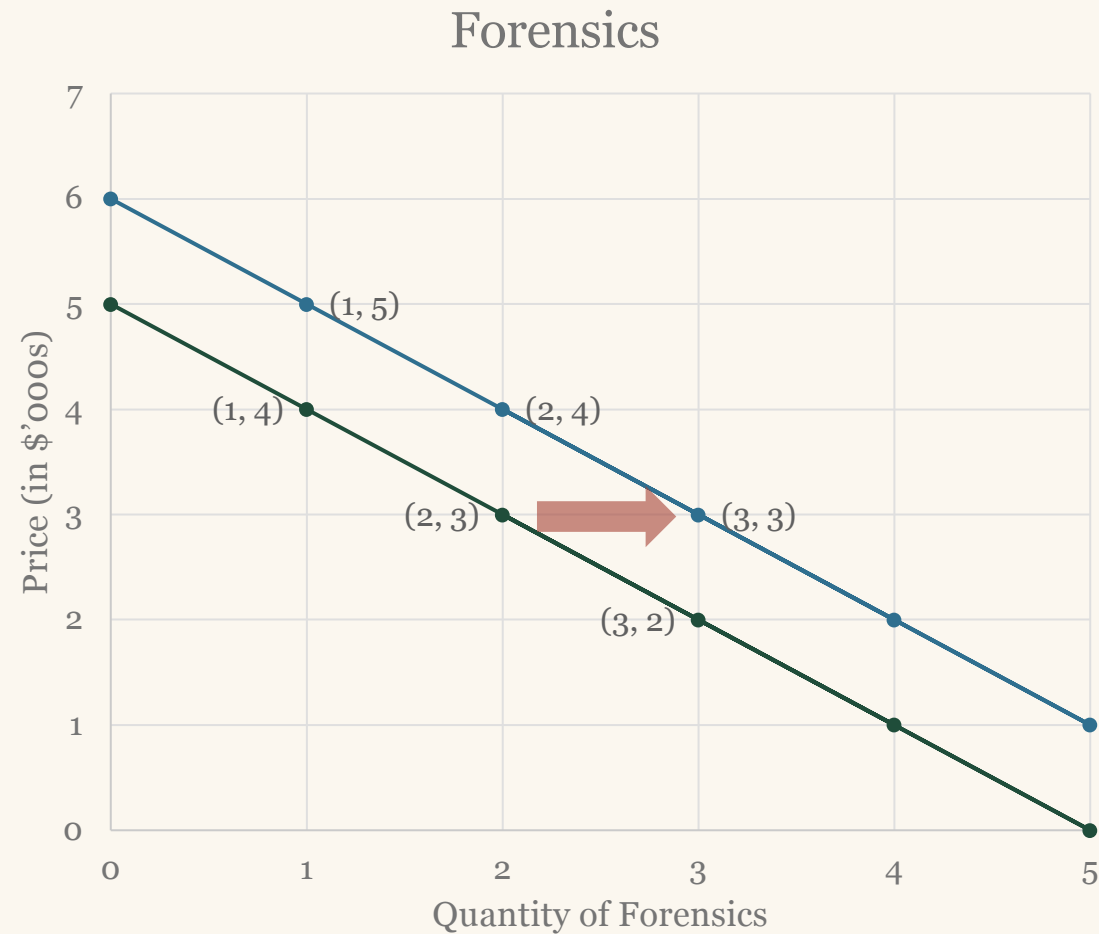
Price of Forensics	Forensics (Old Budget)	Forensics (New Budget)
\$1000	4	5
\$2000	3	4
\$3000	2	3

What Happened When Budget Increased?



Price of Forensics	Forensics (Old Budget)	Forensics (New Budget)
\$1000	4	5
\$2000	3	4
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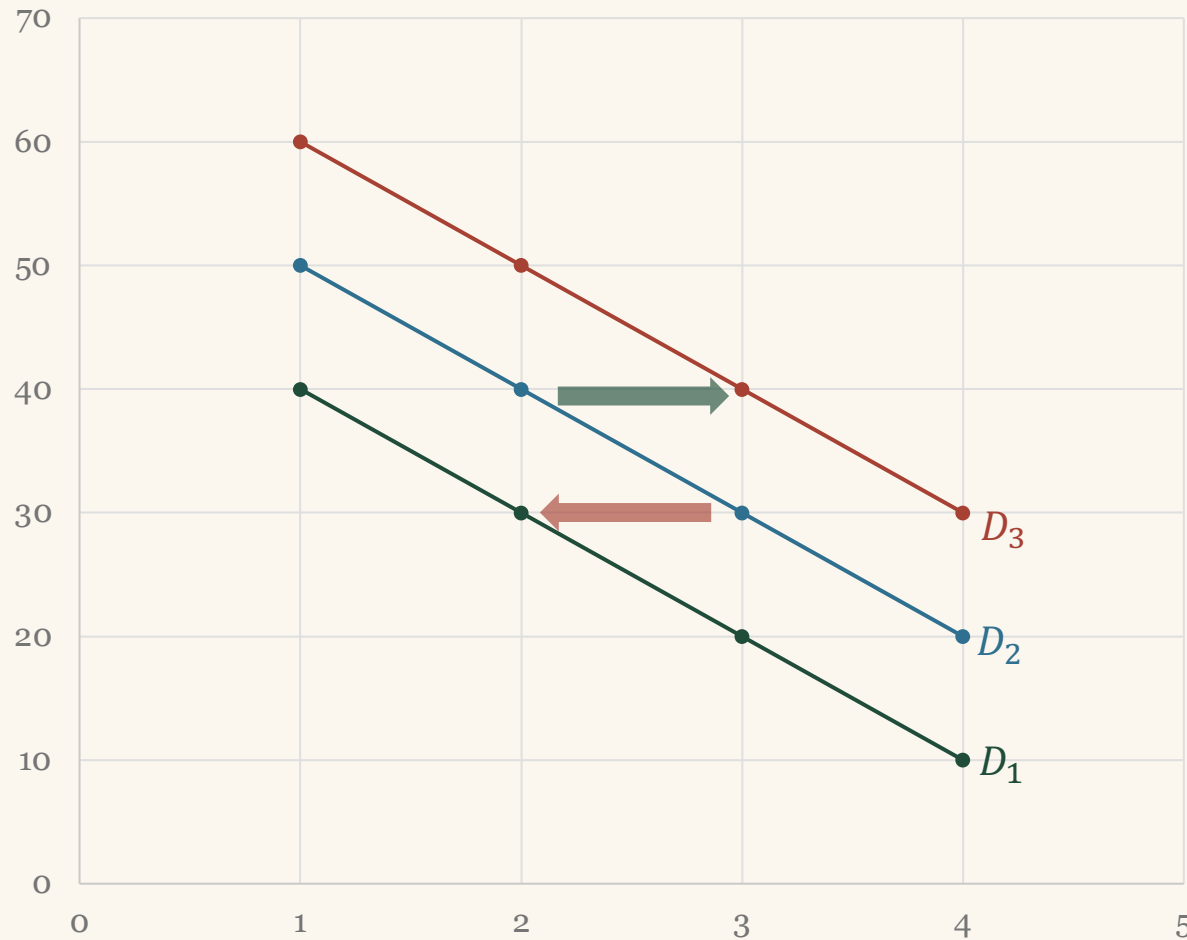
What Happened When Budget Increased?



Price of Forensics	Forensics (Old Budget)	Forensics (New Budget)
\$1000	4	5
\$2000	3	4
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An increase in demand.

Shifting the Demand Curve



$D_2 \rightarrow D_3$: Increase in Demand
 $D_2 \rightarrow D_1$: Decrease in Demand

The quantity demanded changes at every possible price.

Income and Demand

Income changes shift demand.

For a normal good: Income $\uparrow \Rightarrow$ Demand $\uparrow$.

For an inferior good: Income $\uparrow \Rightarrow$ Demand $\downarrow$.

Forensics is a _____ good.



Income and Demand

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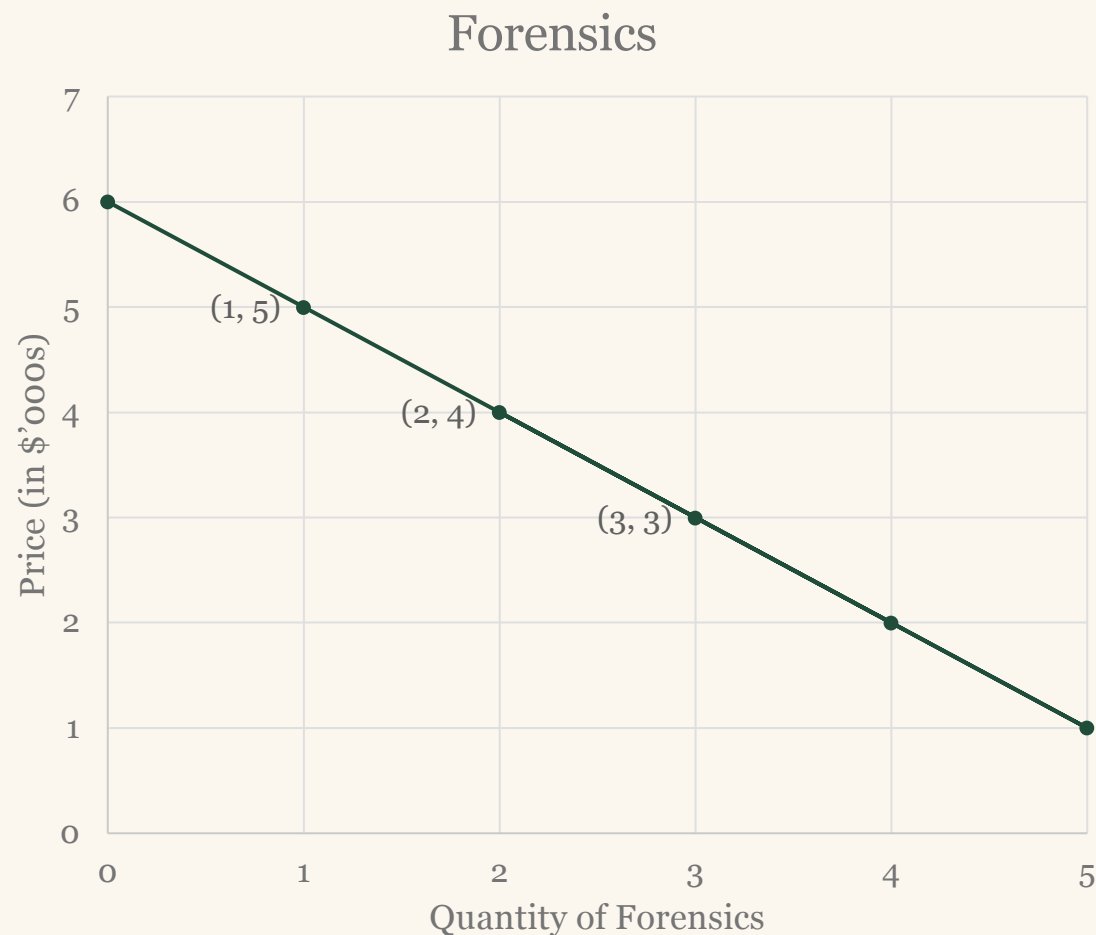
For a normal good: Income $\uparrow \Rightarrow$ Demand $\uparrow$.

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Forensics is a **normal** good.



When Detectives Became Expensive



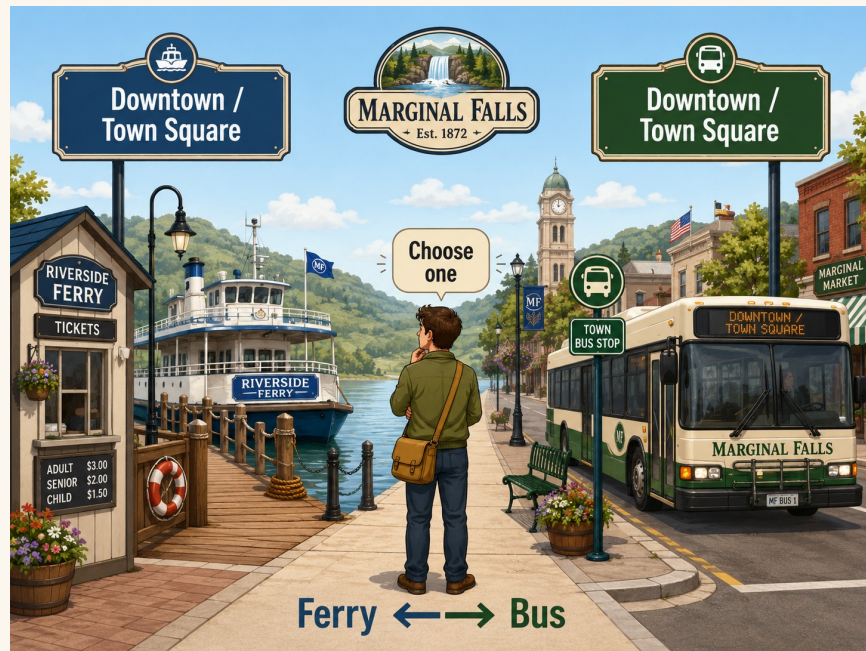
Price of Forensics	Forensics (Old $P_{Detective}$)	Forensics (New $P_{Detective}$)
\$1000	5	4
\$2000	4	3
\$3000	3	2

When Detectives Became Expensive



Price of Forensics	Forensics (Old $P_{Detective}$)	Forensics (New $P_{Detective}$)
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Related Goods: Substitutes and Complements



Substitutes:
can be used in place of each other

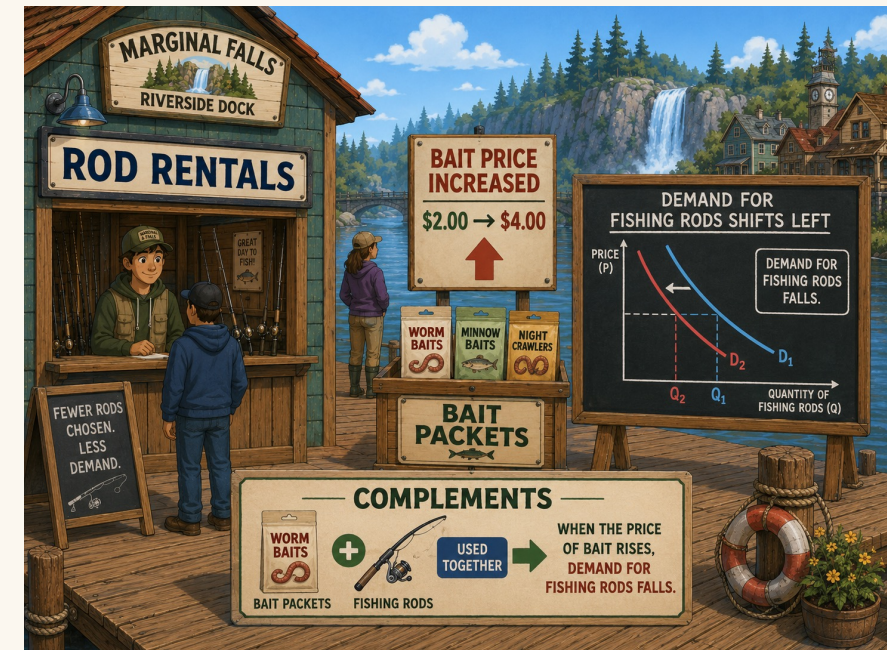


Complements:
are used together

Price of Related Goods



An increase in the bus fares, increase the demand for ferry.



An increase in the price of baits, decreases the demand for fishing rods.

Shift of Demand Curves



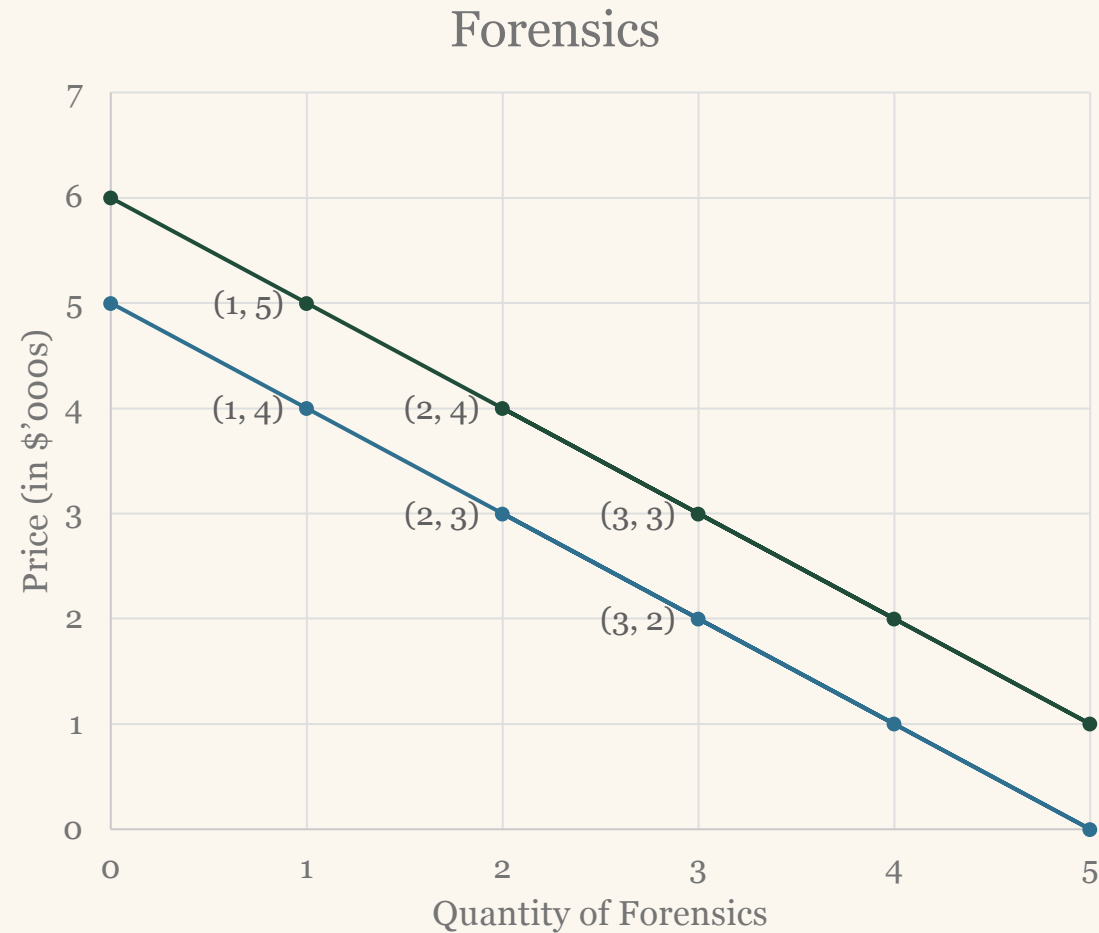
Forensics and Detectives are _____ goods.

Shift of Demand Curves



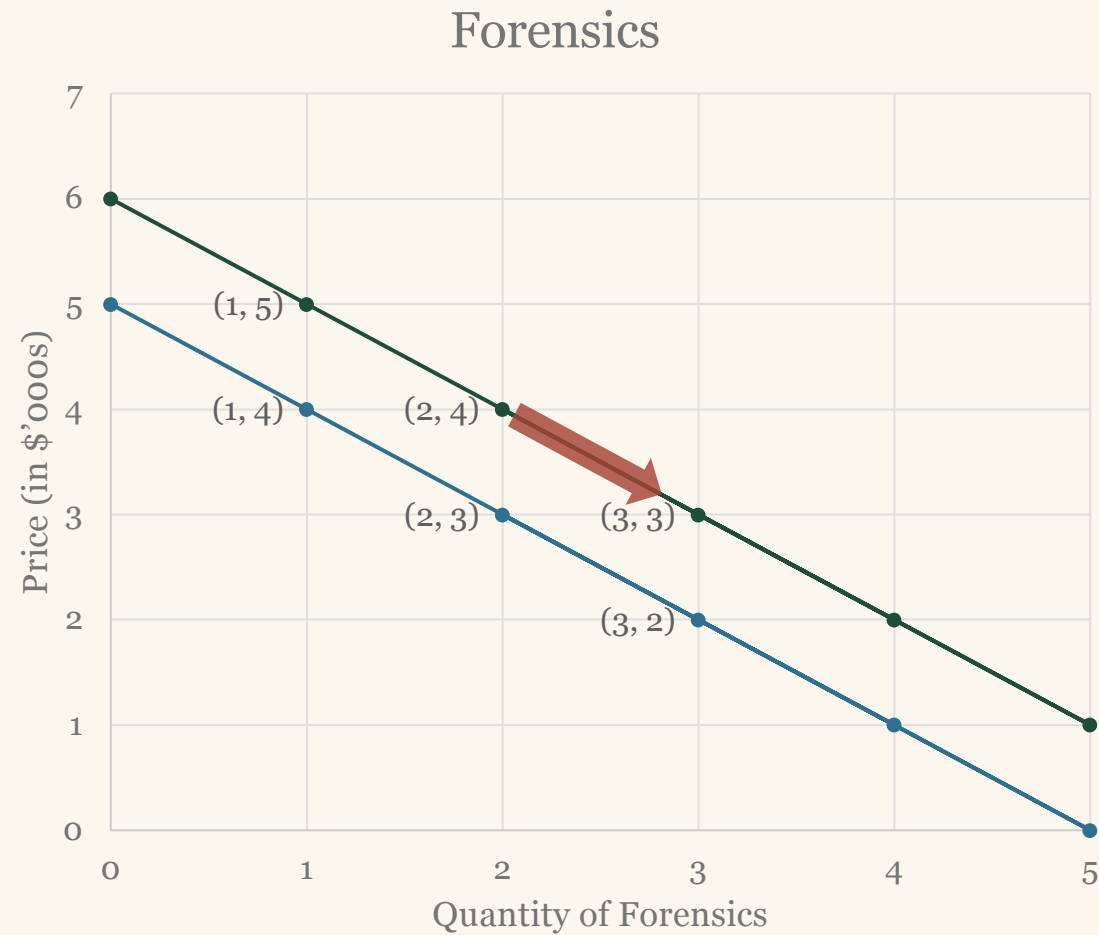
Forensics and Detectives are **complementary** goods.

Shift of Demand Curves



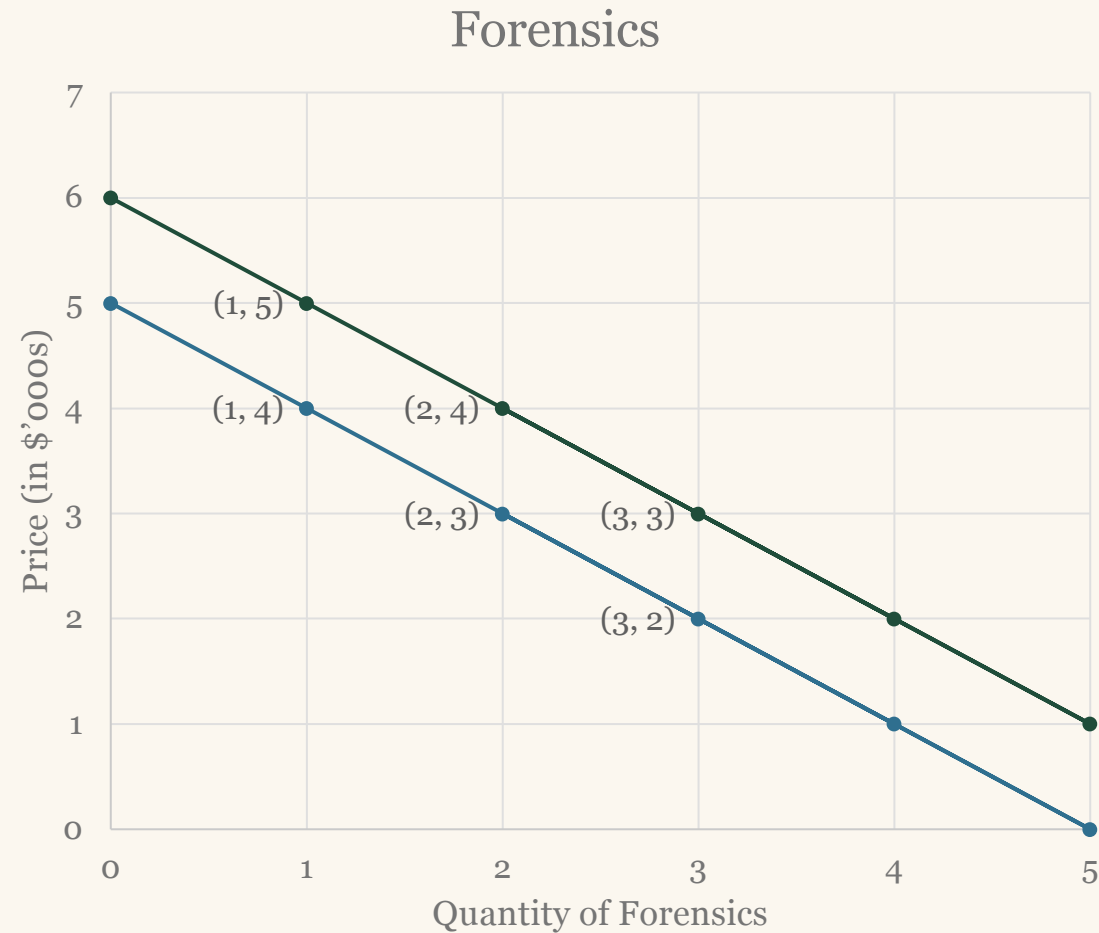
- Own Price change → Movement along demand curve.

Shift of Demand Curves



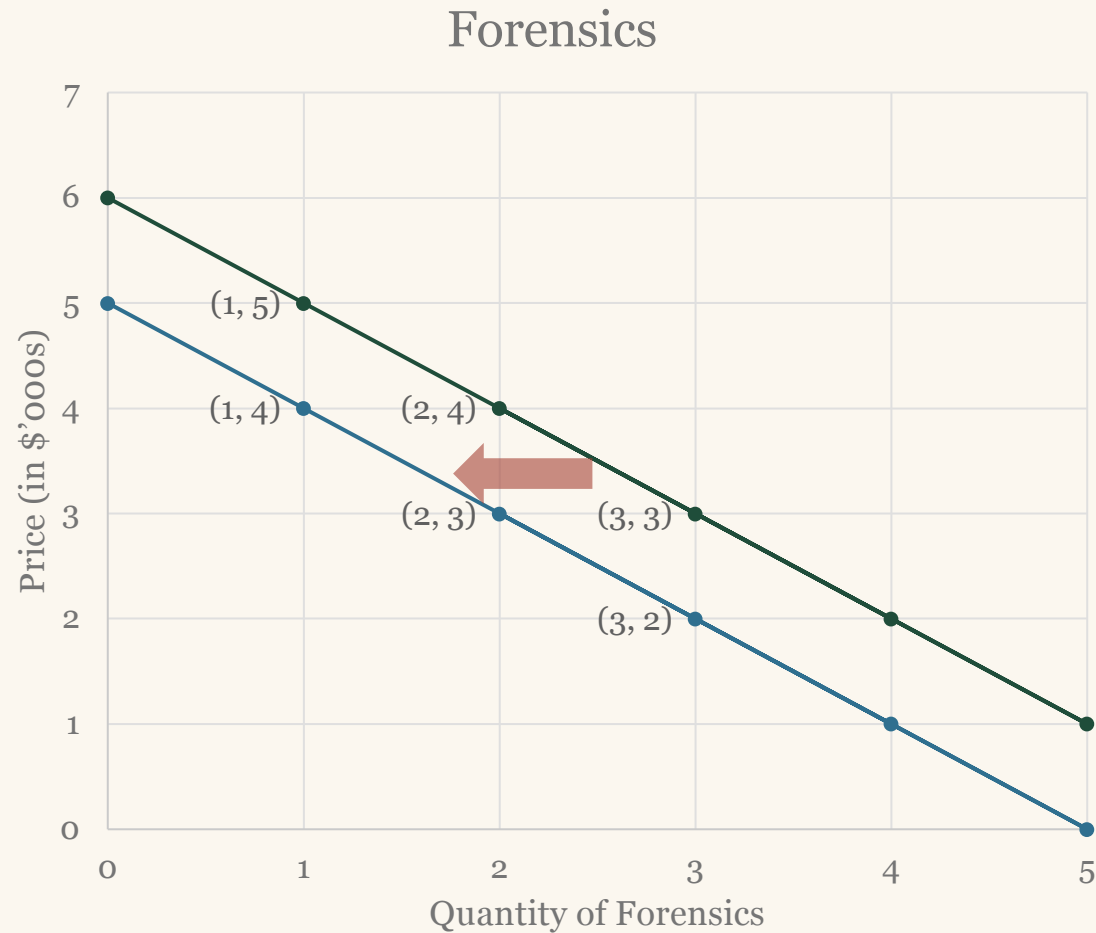
- Own Price change → Movement along demand curve.

Shift of Demand Curves



- Own Price change → Movement along demand curve.
- Other good's price change → Shift of demand curve.

Movement vs Shift



- Own Price change → Movement along demand curve.
- Other good's price change → Shift of demand curve.

From the Chief to the Market

Price of Forensics	Homicide Department	Insurance Fraud Department	River Patrol Department	Market Quantity
\$1,000	4	3	2	
\$2,000	3	2	1	
\$3,000	2	1	0	

- The Chief's **Homicide Department** has one demand curve.
- But Marginal Falls has other departments as well.
- **Insurance Fraud Department** also demands forensics.
- **River Patrol Department** also demands forensics.
- Market demand adds them together.

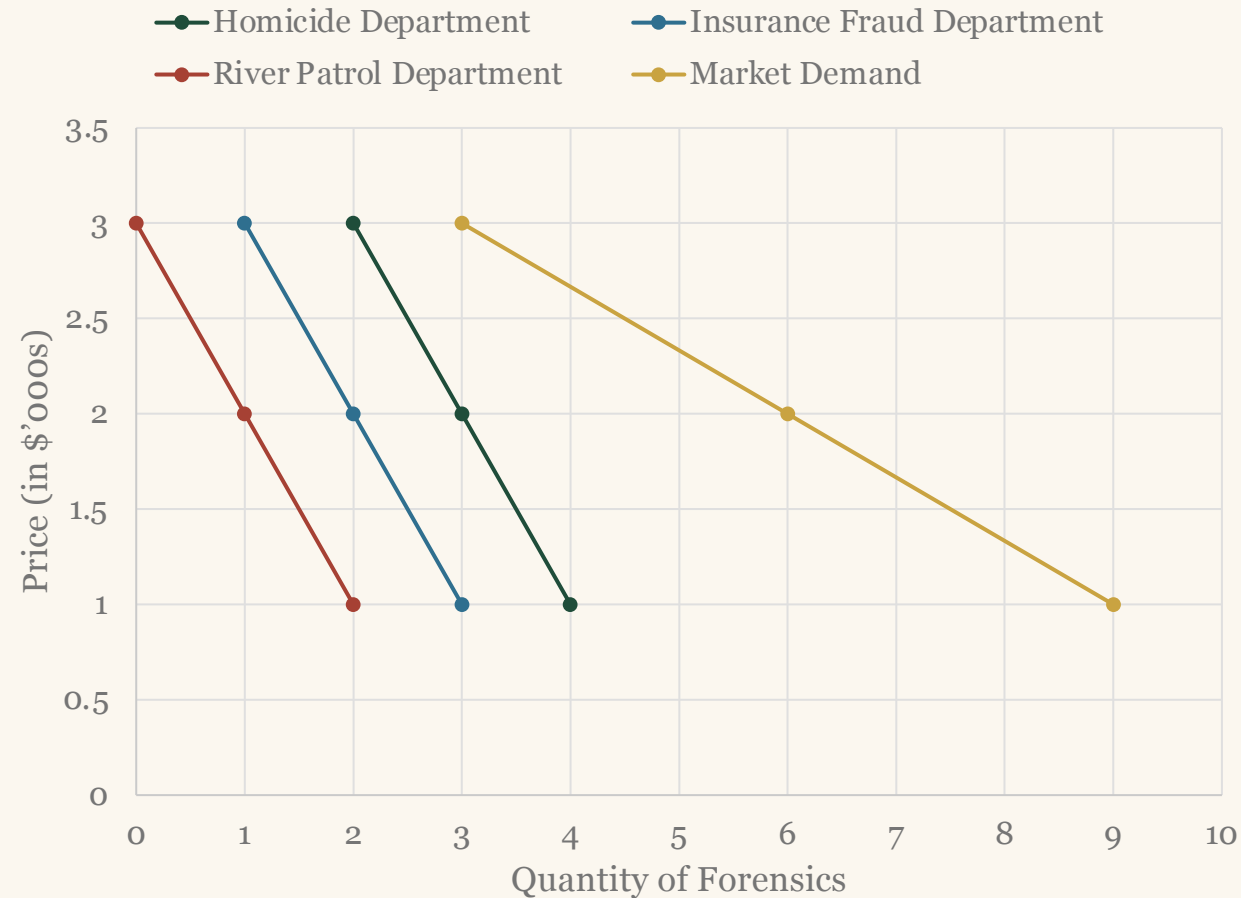
From the Chief to the Market

Price of Forensics	Homicide Department	Insurance Fraud Department	River Patrol Department	Market Quantity
\$1,000	4	3	2	9
\$2,000	3	2	1	6
\$3,000	2	1	0	3

- The Chief's **Homicide Department** has one demand curve.
- But Marginal Falls has other departments as well.
- **Insurance Fraud Department** also demands forensics.
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Demand for Forensics in Marginal Falls

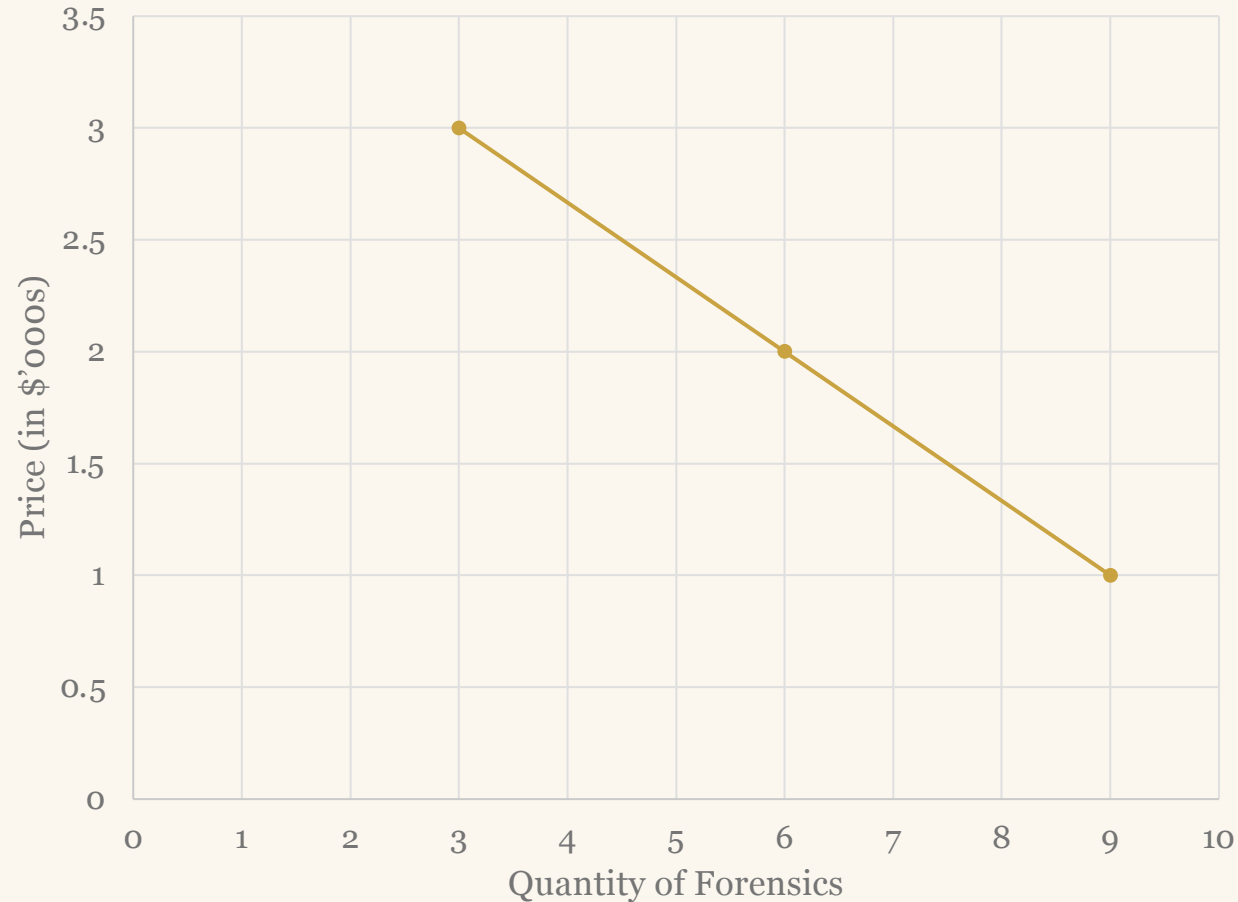
Market Demand



- Market demand is the demand by all buyers of a good or service.
- At each price, market quantity demanded is the sum of individual quantities demanded.

Demand for Forensics in Marginal Falls

Market Demand for Forensics



- Market demand is the demand by all buyers of a good or service.
- At each price, market quantity demanded is the sum of individual quantities demanded.

Up Next in Marginal Falls

Now we know the direction:
Price $\uparrow \rightarrow$ quantity demanded $\downarrow$.

But we still need to know the size of the response.
Does quantity fall a little or a lot?

The answer is **elasticity**.